X-103
NEUTRALISER / FILMING AMINE PACKAGE (ATM OH)
Tank 2 m3 + 2 x 100 % metering pumps

A-101
ATM. OVERHEAD CONDENSER
33.7 MW, 1667 m² bare
2 bay(s), 4 fans x 90 kW

E-115A/B
ATM. OVERHEAD TRIM CONDENSER
6.0 MW, 740 m², TEMA AEU

D-102
ATM. OVERHEAD REFLUX DRUM (3-PHASE)
ID 3.4 m x 12.0 m T/T
boot ID 1.5 m x 3.0 m

P-105A/B
OVERHEAD SOUR WATER
14 m³/h @ 51 m
5.5 kW, API 610 OH2

P-103A/B
ATM. REFLUX
159 m³/h @ 102 m
55 kW, API 610 OH2

P-104A/B
UNSTABILISED NAPHTHA
180 m³/h @ 232 m
160 kW, API 610 OH2

E-114
STABILISER FEED / BOTTOMS
3.4 MW, 117 m², TEMA AES

C-105
NAPHTHA STABILISER (DEBUTANISER)
ID 2.1 m x 18.0 m T/T
19 valve trays

E-116
STABILISER REBOILER (HP STEAM)
8.5 MW, 190 m², TEMA BKT

D-105
STABILISER REFLUX DRUM
ID 1.7 m x 5.5 m T/T
boot ID 0.6 m x 1.5 m

A-106
STABILISER OVERHEAD CONDENSER
2.1 MW, 214 m² bare
1 bay(s), 2 fans x 11 kW

P-115A/B
STABILISER REFLUX / LPG
39 m³/h @ 154 m
18.5 kW, API 610 OH2

C-106
NAPHTHA SPLITTER
ID 3.1 m x 29.5 m T/T
38 valve trays

P-117A/B
HEAVY NAPHTHA PRODUCT
142 m³/h @ 98 m
45 kW, API 610 OH2

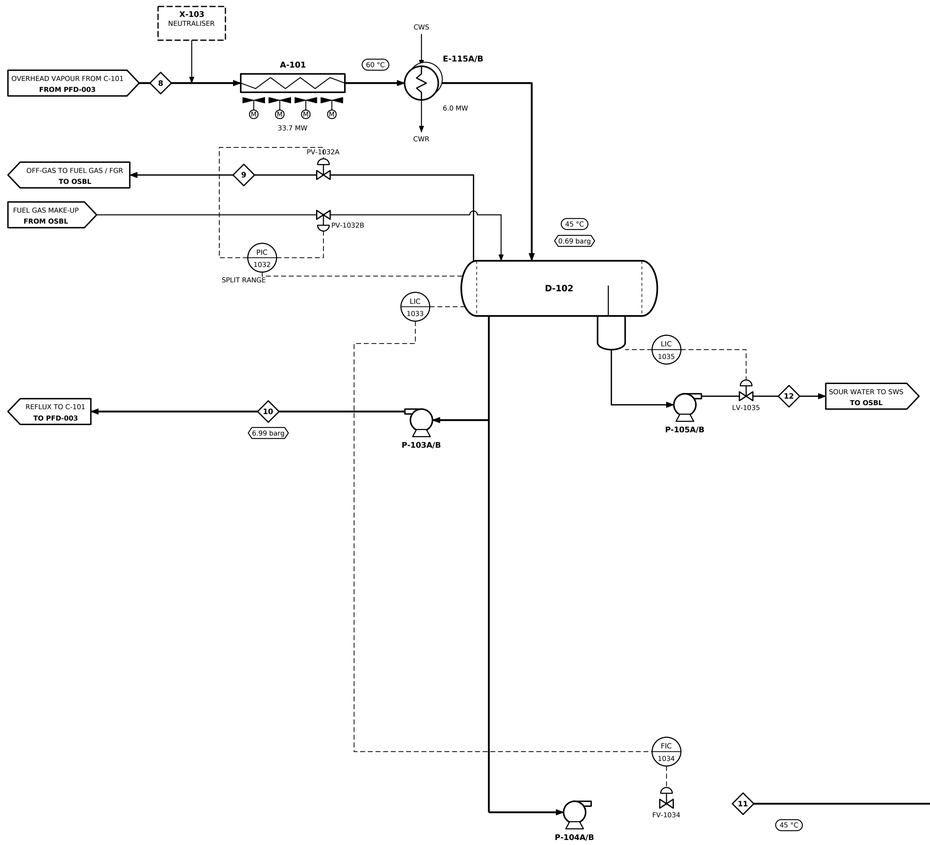
E-117
SPLITTER REBOILER (MP STEAM)
6.8 MW, 225 m², TEMA BXM

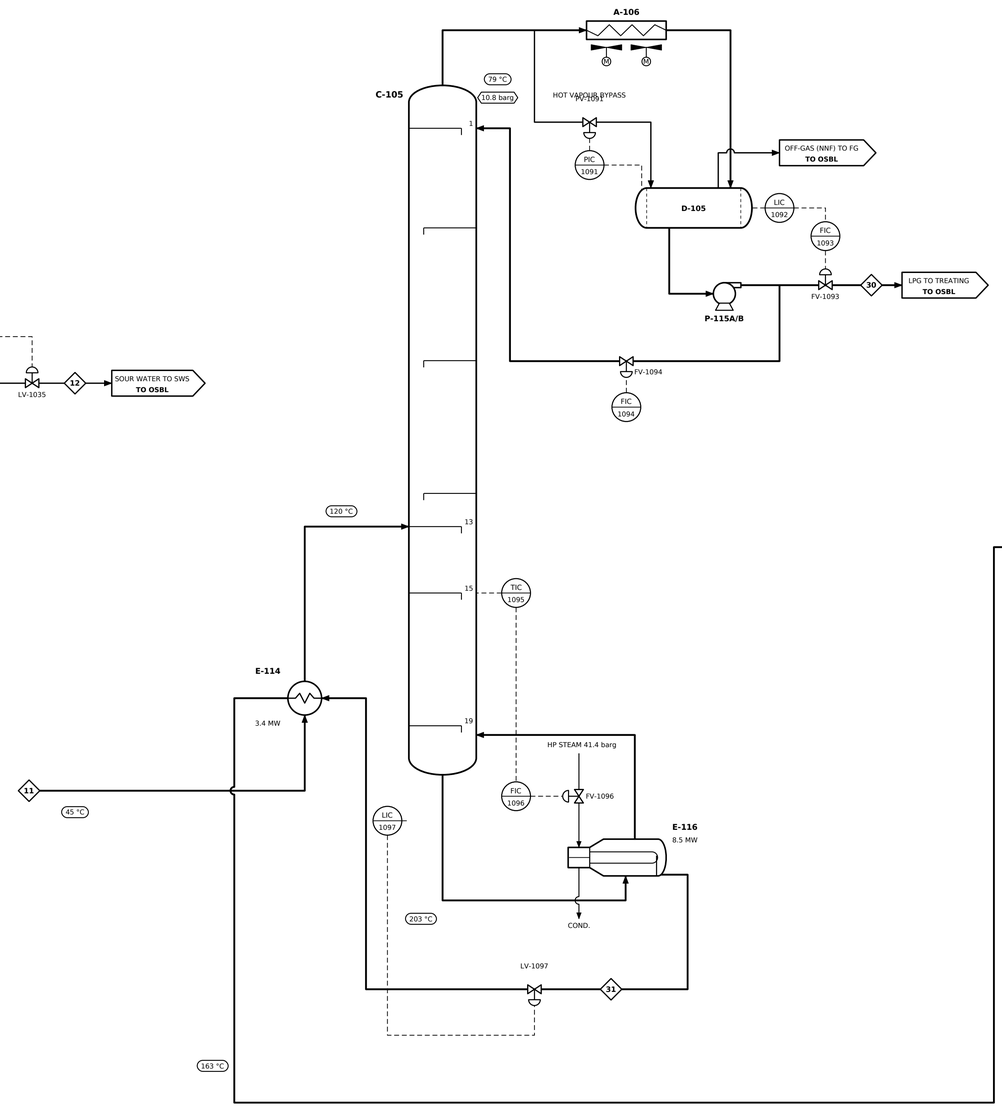
A-107
SPLITTER OVERHEAD CONDENSER
9.6 MW, 881 m² bare
1 bay(s), 2 fans x 45 kW

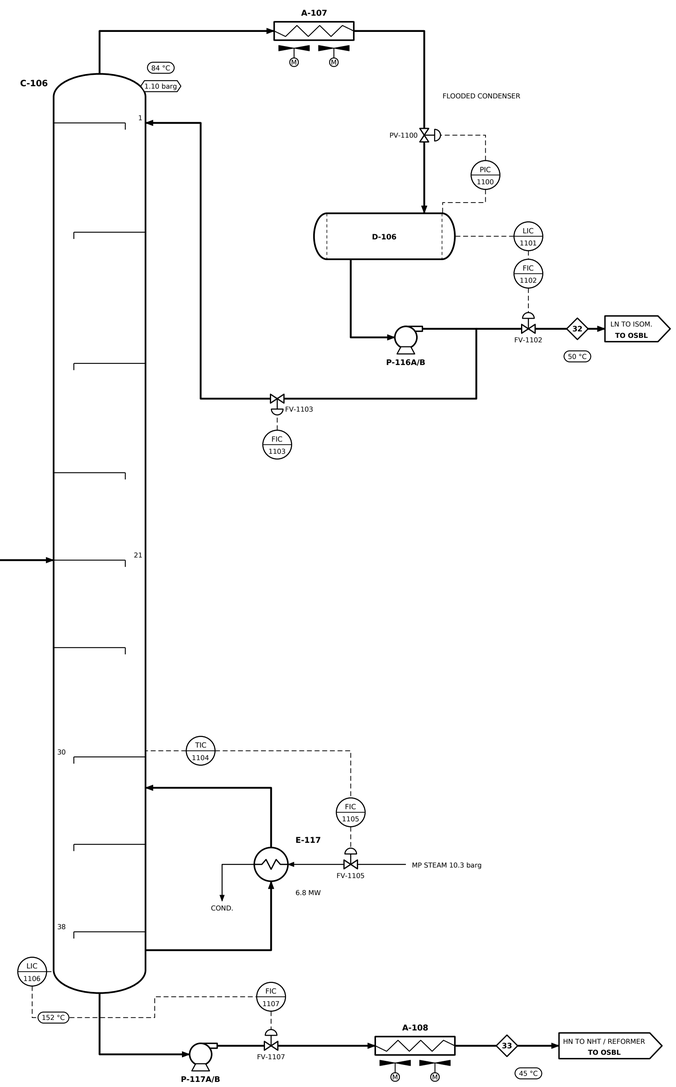
D-106
SPLITTER REFLUX DRUM
ID 2.7 m x 8.5 m T/T

P-116A/B
SPLITTER REFLUX / LN
146 m³/h @ 111 m
55 kW, API 610 OH2

A-108
HEAVY NAPHTHA PRODUCT COOLER
5.8 MW, 401 m² bare
1 bay(s), 2 fans x 30 kW







LEGEND

- ◇00◇ H&MB STREAM No.
- (000 °C) TEMPERATURE
- (0.0 barg) PRESSURE
- (FIC 000) DCS CONTROLLER
- INSTRUMENT SIGNAL
- CONTROL VALVE
- SERVICE TO PFD-00X — OFF-SHEET CONNECTOR
- MAIN PROCESS
- SECONDARY PROCESS
- UTILITY
- FLUE GAS / AIR

NOTES

1. D-102 pressure by split range: PV-1032A off-gas to FG/FGR, PV-1032B fuel gas make-up.
2. C-105 pressure by hot-vapour bypass PV-1091; C-106 by flooded condenser PV-1100.
3. Reboiler duties set by sensitive-tray temperature cascades (TIC-1095, TIC-1104).
4. Stream numbers (diamonds) refer to H&MB CFU-000-PR-HMB-001; data from design case, normal operation.
5. Pressures bar(g) unless noted. Pumps A/B are 2 x 100 %; only duty pump shown.

STREAM No.	◇8◇	◇9◇	◇10◇	◇11◇	◇12◇	◇30◇	◇31◇	◇32◇	◇33◇
DESCRIPTION	Column overhead vapour	Overhead drum off-gas (NNF)	Reflux	Unstabilised naphtha	Overhead sour water	LPG product	Stabilised naphtha	Light naphtha	Heavy naphtha
PHASE	VAPOUR	VAPOUR	LIQUID	LIQUID	WATER	LIQUID	LIQUID	LIQUID	LIQUID
TOTAL FLOW kg/h	229,224	0	101,310	114,910	13,004	6,468	108,442	27,607	80,835
TEMPERATURE °C	134	45	45	45	45	45	203	50	45
PRESSURE barg	1.2	0.7	7.0	14.0	5.0	17.0	11.3	7.0	7.0
STD LIQ. FLOW BPSD	45,042	-	21,105	23,938	-	1,743	22,195	5,920	16,275
VAPOUR FRACTION (mass)	1	-	0	0	-	0	0	0	0
MOLECULAR WEIGHT	97.9	-	97.9	97.9	-	54.2	102.9	81.0	113.3

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)

GENERIC REFINERY - GULF COAST
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT

PROCESS FLOW DIAGRAM
ATMOSPHERIC OVERHEAD SYSTEM, STABILISER & NAPHTHA SPLITTER

DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	4 OF 6

DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02

DRAWING NUMBER: **CFU-100-PR-PFD-004**
REV: **A**